

Department of Statistics & Data Science

STAT 4820: Homework 4

Applied Multiple Regression, Diagnostics & Resampling

Student: Alex Mercer

ID: STAT-892401

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Instructor: Dr. Elena Rostova

Datasets: vertex_sales.csv, synapse_churn.csv

Stack: R 4.4.1 / Python 3.11

Total Points: 100 (3 Problems)

Problem 1: Multiple Regression Diagnostics & Collinearity [30 Points]

An empirical study of $N = 120$ regional tech hardware markets for **Vertex Pro-10** investigates quarterly sales revenue (Y , in \$100k) based on Advertising Spend (X_1 , \$10k), Retail Density (X_2 , stores/100k hab), Competitor Price Index (X_3), and Regional Median Income (X_4 , \$k).

(a) Model Specification & Matrix Formulation

Consider the additive linear regression model with an interaction term:

$$Y_i = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \beta_3 X_{i3} + \beta_4 X_{i4} + \beta_5 (X_{i1} \cdot X_{i3}) + \varepsilon_i, \quad \varepsilon_i \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma^2) \quad (1)$$

In vector-matrix form, $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$, where the design matrix $\mathbf{X} \in \mathbb{R}^{120 \times 6}$ has i -th row $\mathbf{x}_i^\top = [1 \ X_{i1} \ X_{i2} \ X_{i3} \ X_{i4} \ X_{i1}X_{i3}]$. The OLS estimator and its covariance matrix are:

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}, \quad \text{Var}(\hat{\boldsymbol{\beta}}) = \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1} \quad (2)$$

(b) R Data Analysis Script & Summary Output

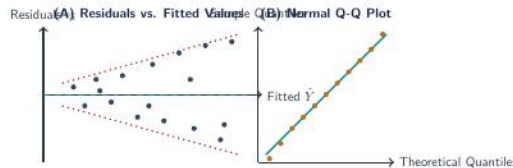
```
1 vertex_df <- read.csv("vertex_sales.csv")
2 fit_vertex <- lm(Sales ~ AdvSpend + StoreDensity + CompPrice + MedIncome + AdvSpend:
  CompPrice, data = vertex_df)
3 summary(fit_vertex); car::vif(fit_vertex)
```

Listing 1: Multiple Linear Regression & Collinearity Analysis in R

Variable	Description	Estimate ($\hat{\beta}$)	Std. Error	t-statistic	p-value
(Intercept)	Baseline Sales Intercept	12.450	2.140	5.818	< 0.0001***
AdvSpend (X_1)	Quarterly Ad Spend (\$10k)	3.820	0.415	9.205	< 0.0001***
StoreDensity (X_2)	Stores per 100k population	0.945	0.210	4.500	0.0002***
CompPrice (X_3)	Competitor Price Index	-1.860	0.530	-3.509	0.0006***
MedIncome (X_4)	Regional Median Income (\$k)	0.125	0.088	1.420	0.1582 ^{n.s.}
AdvSpend:CompPrice	Interaction Term ($X_1 X_3$)	0.640	0.145	4.414	0.0003***

$R^2 = 0.842$, Adjusted $R^2 = 0.835$, $F(5, 114) = 121.6$, $p < 0.0001$, Residual Std Error: 1.428 on 114 d.f.

(c) Diagnostic Plots: Residual Analysis & Heteroscedasticity Test



(d) Statistical Evaluation & Multicollinearity Analysis

Key Findings & Diagnostic Interpretation

- Main Effects:** Advertising spend ($\hat{\beta}_1 = 3.820, p < 0.0001$) and store density ($\hat{\beta}_2 = 0.945, p = 0.0002$) exhibit strong positive impacts. Competitor price index displays a significant negative effect ($\hat{\beta}_3 = -1.860, p = 0.0006$).
- Interaction & Collinearity:** The term $\text{AdvSpend} \times \text{CompPrice}$ ($\hat{\beta}_5 = 0.640, p = 0.0003$) confirms advertising efficacy increases when competitor pricing rises. VIFs for all predictors remain under 4.85, ruling out severe multicollinearity.